



Hoof lesion detection in lactating Holsteins: Part II. Development and on-farm validation of a predictive classification model using autonomous camera system data compared to passive surveillance

D. Swartz,^{1*}† A. Rendahl,² and G. Cramer¹

¹University of Minnesota, Department of Veterinary Population Medicine, St. Paul, MN 55108

²University of Minnesota, Department of Veterinary and Biomedical Sciences, St. Paul, MN 55108

ABSTRACT

Hoof lesion detection remains a challenge in lameness management on dairy farms. Recent studies have proposed locomotion score (LS)-based thresholds using an autonomous camera system (AUTO), which generates BCS and LS. However, these thresholds have limited ability to distinguish affected cows. In the first phase, the primary objective was to develop and tune classification models optimized for the $F_{0.5}$ scores, which emphasize positive predictive value (PPV) over sensitivity. In the second phase, which was a prospective, live implementation design, the primary objective was to evaluate the positive classifications generated by the top-performing approach identified in phase 1, with a secondary objective of comparing its PPV to that of the farm's existing passive surveillance approach, defined as on-farm identification of cows for hoof trimming independent of structured locomotion scoring. In phase 1, a total of 511 Holstein cows between 8 and 100 DIM with no typically treated hoof lesions and no more than one prior noninfectious lesion were enrolled from 2 sites. Cows were classified based on hoof trimming outcomes as typically treated (TT), which included hoof lesions typically requiring treatment (e.g., digital dermatitis, sole ulcer, white line disease), or as typically not treated (TNT), which included cows without hoof lesions or without lesions typically requiring treatment. Five machine learning algorithms incorporating AUTO BCS and LS, along with health and milk production data, were tuned and evaluated across multiple feature sets using cross-validation. For the on-farm evaluation, cows similar to the enrollment criteria were processed by the model, and those classified as positive were trimmed by the on-site hoof trimmer. The PPV was calculated against

on-site hoof trimmer-reported nonhealthy outcomes and compared with PPV from cows flagged through passive surveillance using the same subset. During development, the top-performing model was a random forest model incorporating AUTO-derived BCS and LS features from the 14-d period before hoof trimming, selected based on the highest $F_{0.5}$ score (59.1), with PPV values of 100% (95% CI: 100%, 100%) in training and 92.3% (95% CI: 76.4%, 100%) in testing. During live implementation, this random forest model yielded a lower PPV of 13.7% (22/187; 95% CI: 9.9%–17%). Meanwhile, passive surveillance yielded a greater PPV of 48.6% (34/70; 95% CI: 40%–57.9%). Both passive surveillance and random forest identified a total of 78 cows with an outcome at hoof trimming. Of these, 12.8% (10/78) were flagged by both random forest and passive surveillance, 39.7% (31/78) by the random forest model only, and 47.4% (37/78) by passive surveillance only. Although the random forest model demonstrated lower PPV than passive surveillance, each approach identified cows that were not detected by the other. Although the model resulted in potentially more unnecessary trims, it also flagged cows that would have otherwise gone undetected. These findings highlight the potential complementary value of the model for hoof lesion detection; however, increased unnecessary trims may raise costs or occupy trimming capacity, delaying treatment for cows requiring intervention.

Key words: lameness, dairy cow, precision livestock farming

INTRODUCTION

Lameness is a clinical sign of underlying conditions that impair locomotion (Archer et al., 2010), with hoof lesions being the primary cause. Cows affected by hoof lesions are more likely to exhibit lameness (van Huysteen et al., 2020), with the most prevalent hoof lesions in North American dairy cattle being digital dermatitis, sole ulcers, and white line disease (Cramer et al., 2008; DeFraen et al., 2013; Solano et al., 2016). Lameness has

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*Corresponding author: swartzdrew@yahoo.com

†Current address: Department of Animal, Dairy and Veterinary Sciences, Utah State University, Logan, UT 84322.

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

serious implications for the dairy industry, contributing to poor animal welfare (Whay and Shearer, 2017), economic losses (Cha et al., 2010; Ózsvári, 2017), and increased greenhouse gas emissions (Mostert et al., 2018). With lameness prevalence estimates ranging from 10% to 55% in North America (von Keyserlingk et al., 2012; Cook et al., 2016; Adams et al., 2017), improving the detection and management of hoof lesions is essential to reducing the burden of lameness.

Given that the incidence of new hoof lesions is low (Wynands et al., 2022; Stoddard et al., 2025), the reason prevalence is high is that a large portion of lameness cases are chronic (Randall et al., 2018). This distinction is particularly relevant to the present study, which focuses on identifying new hoof lesions before the onset of chronicity, highlighting the need for improved detection methods to support effective lameness management. Although passive surveillance, defined as the routine, nonsystematic identification of cows for hoof trimming during daily farm activities without the use of structured locomotion scoring, remains the predominant strategy on dairy farms, consistent visual locomotion scoring has been demonstrated to improve lameness management (Leach et al., 2012; Thomas et al., 2015; Wilson et al., 2022). However, visual locomotion scoring has limited sensitivity for detecting hoof lesions (Logan et al., 2024), and prioritizing sensitivity without considering lesion presence can lead to unnecessary interventions. Stoddard and Cramer (2024) found that trimming cows without hoof lesions led to short-term reductions in milk yield in some herds, illustrating the costs of interventions with limited preventive value. For example, Hernandez et al. (2007) reported limited benefits of mid-lactation trims in reducing the subsequent incidence of lameness and hoof lesions. Further supporting this, Van Hertem et al. (2014) reported that the effects of trimming on milk yield varied by parity and lactation stage, suggesting that hoof trimming selection should be tailored to individual cow characteristics to optimize outcomes rather than relying solely on routine preventive schedules.

Because many producers already conduct 2 routine trimmings per lactation (USDA, 2016; Swartz, 2025), the value of adopting an automated detection system depends on its ability to accurately identify only those cows that require trimming. If these systems frequently flag cows without lesions, they replicate the same disruptions and labor costs associated with routine trimming, while also incurring the added expense of the technology. O'Leary et al. (2020) indicated that producers' trust in a system is most closely tied to its positive predictive value (PPV), as high false-positive rates can diminish confidence in a tool's usefulness. Similarly, Van Nuffel et al. (2015) suggest that repeated evaluation of healthy animals due to false alerts increases labor demands and undermines

the perceived value of detection technologies. Therefore, detection systems intended for on-farm use should prioritize PPV to promote producer confidence.

Multiple studies have evaluated a commercially available autonomous camera-based system (AUTO; Cattle-Eye Ltd., Belfast, UK) as a tool for identifying cows with hoof lesions (Anagnostopoulos et al., 2023; Swartz et al., 2024; Siachos et al., 2025). In a previous external validation study (Swartz and Cramer, 2026), we assessed the performance of recommended AUTO-derived locomotion score (LS) thresholds (Anagnostopoulos et al., 2023; Siachos et al., 2025) and found that they yielded poor PPV when applied to cows with one or fewer previous hoof lesions. To support decision-making in contexts where false positives carry meaningful consequences, Goutte and Gaussier (2005) showed that adjusting the β parameter in the F β -score enables users to tailor the balance between sensitivity and PPV. When using 0.5 for β (F_{0.5}), the F_{0.5} score allows greater emphasis on PPV while still accounting for sensitivity (Saito and Rehmsmeier, 2015) making it a useful way to evaluate hoof lesion detection methods.

Given the limited effectiveness of current hoof lesion detection methods, particularly with respect to PPV, this study aimed to evaluate whether machine learning models optimized to improve positive predictions could enhance identification of cows with hoof lesions. In the first phase, the primary objective was to develop and tune classification models optimized for the F_{0.5} scores. In the second phase, which represented a live implementation setting, the primary objective was to evaluate the positive classifications generated by the top-performing approach evaluated in phase 1 with a secondary objective of comparing the positive predictive value to that of the current farm approach.

MATERIALS AND METHODS

Phase 1: Model Development

Study Design and Housing and Cow Characteristics. The study design and housing characteristics of the animals enrolled in this study are fully described in Swartz and Cramer (2026). Briefly, this case-cohort study (O'Brien et al., 2022) was conducted from July to November 2023 on a commercial dairy operation in Minnesota with 2 sites differing by parity. Site 1 housed multiparous cows in stalls with a mix of deep-bedded and mattress bases, and site 2 housed primiparous cows in mattress-based stalls; both were bedded with manure solids. All barns featured grooved concrete flooring and a 3× daily milking schedule. Cows remained indoors throughout their lactation. All cows calved at site 1, with primiparous cows transferred to site 2 at 10 ± 3 DIM

Table 1. Identified conditions (diagnosis) and their descriptions, along with study eligibility and outcome classification groups (lesion not requiring treatment [LNRT], lesion requiring treatment [LRT], or pain); adapted from Egger-Danner et al. (2020)

Diagnosis	Zone ¹	Definition	Enrollment eligible ²	Outcome classification
Corkscrew claw	7, 8	Any torsion of either the outer or inner hoof; the dorsal edge of the wall deviates from a straight line	Yes	LNRT
Heel erosion	6	Eroded heel horn that involved a flap or severely degraded horn tissue	Yes	LNRT
Horizontal fissure	7, 8	Horizontal crack in the hoof wall	Yes	LNRT
Interdigital hyperplasia	0	Interdigital growth of fibrous tissue	Yes	LNRT
M1 ³	10	An ulcerative lesion less than 2 cm in diameter	Yes	LNRT
M2 ³	10	An ulcerative lesion greater than or equal to 2 cm in diameter	No	LRT
M3 ³	10	A scab-like appearance covering where a M1 or M2 would have been located	Yes	LNRT
M4 ³	10	Raised skin with hyperkeratotic/proliferative growths	Yes	LNRT
M4.1 ³	10	An ulcerative lesion that occurs on a M4c growth that is less than 2 cm in diameter	Yes	LNRT
Swollen	Non-hoof	Any swelling occurring on the P1, P2, hock	No	LNRT
Hock or knee	Non-hoof	Any ulcerative and bleeding lesions occurring on the hock or knee that are not swollen	Yes	LNRT
Other ⁴	Unknown	Other outcomes recorded by the site hoof trimmers:	NA ⁵	LRT
Sole hemorrhage	4	Diffused or circumscribed red or yellow discoloration of the sole	Yes	LNRT
Sole ulcer	4	A defect in sole horn that exposes corium	No	LRT
Toe lesion	1, 2, 5	Combination of thin soles (horn that yields with digital pressure) and toe ulcer (exposure of corium through the sole horn)	No	LRT
White line hemorrhage	3	Diffused or circumscribed red or yellow discoloration of the white line	Yes	LNRT
White line disease	3	Separation of the white line with or without purulent exudation	No	LRT
Pain	Unknown	Any cow eliciting a withdrawal reflex with hoof testers	No	Pain

¹Zone of which the diagnosis occurred in according to the 13-zone scoring system (Shearer et al., 2004). Non-hoof = diagnosis location not on the hoof; unknown = no reported zone for diagnosis.

²Eligibility to continue if this diagnosis was found at the enrollment trimming.

³Stages of digital dermatitis (Berry et al., 2010).

⁴Consists of different lesions reported by the site hoof trimmers: interdigital hyperplasia, periople ulcer, non-ulcer, other, puncture, vertical fissure, axial fissure, corkscrew.

⁵NA = not applicable, because the site hoof trimmers did not perform the enrollment evaluations.

unless delayed by a health event. A sample size of 500 Holstein cows was used which was determined based on available funding and study design recommendations provided by the sponsor (CattleEye Ltd., Belfast, UK). All protocols involving animals in this study were approved by the University of Minnesota Institutional Animal Care and Use Committee (St. Paul, MN), protocol number 2206-40148A.

Enrollment and Follow-Up Evaluation. The enrollment and follow-up procedures are fully described in Swartz and Cramer (2026). Briefly, cows were eligible if they were Holstein, between 8 and 100 DIM, no longer in the fresh pen, and had no more than one prior hoof lesion. Cows were brought to the hoof trimming chute and received a superficial trim (removal of a superficial sliver of horn), performed primarily by a single study personnel member with occasional assistance from a second trained personnel member, followed by a visual hoof inspection using the 13-zone system, alongside hock and knee evaluations. Hoof testers were applied to assess pain response, and lesions were recorded via rugged touchscreen tablet (Hoof Supervisor, Dresser, WI). At enrollment, cows with diagnoses of digital dermatitis (**M2**; an ulcerative lesion ≥ 2 cm in diameter), swollen first phalanx (**P1**) or second phalanx (**P2**), hock or knee lesions with swell-

ing (lesions without swelling were eligible to continue), sole ulcer, toe ulcer, white line disease, or those that elicited a withdrawal reflex were treated and excluded from the study. Lesion definitions and exclusion criteria are provided in Table 1. Follow-up evaluations occurred 100 to 120 d after enrollment, which included a routine hoof trim and an evaluation using the same procedures described at enrollment.

Cow and Event Data. Cow-level data were extracted from the operation's herd management software and downloaded as .csv files. The daily data included individual cow records for daily milk production, DIM, and lactation number. Additionally, a file containing recorded health events and hoof trimming history was downloaded. Health events were recorded by site staff following each site's standard operating procedures. Hoof trimming history included records of trimmings performed by the site-specific hoof trimmers ($n = 2$), both of whom had been previously trained and calibrated by study personnel (G.C.) in lesion identification and recording.

Body Condition and Locomotion Camera System Data. Additionally, AUTO data were collected daily, starting the day after enrollment through the day before the follow-up evaluation. The AUTO system accurately identifies individual cows (Swartz et al., 2025) gener-

ates LS (Anagnostopoulos et al., 2023), and assigns BCS (Siachos et al., 2024). The system uses an overhead 2-dimensional camera (~4 m high) installed in the return alley of the milking parlor to capture video as cows walk 3 to 4 m beneath it. Cow identification was achieved through radio-frequency identification tags and visual recognition. To reduce congestion and ensure daily recordings, cameras were scheduled to run from 1700 h to 0030 h (considered 1 d) at site 1 and from 1300 h to 2100 h at site 2 each day.

Video recordings are transmitted to Amazon Web Services, where algorithms assign 2 key scores: BCS, evaluated on a 5-point scale in 0.25-unit increments (1 = most under-conditioned, 5 = most over-conditioned; Ferguson et al., 1994), and LS, assigned on a scale of 0 to 100 in 1-point increments. The study personnel received LS and BCS for enrolled cows directly from the AUTO personnel, who at the time of data transfer were blinded to the results of the follow-up evaluations.

Data Management and Analysis. All data processing and modeling were performed in R (v4.4.2; R Core Team, 2024) using RStudio (v.2024.9.1.394; Posit team, 2024).

Creating Outcomes of Interest. As detailed in Swartz and Cramer (2026), hoof lesion diagnosis, definitions, and initial classifications of either lesion not requiring treatment (LNRT; which includes cows without hoof lesions or those with lesions not requiring intervention beyond routine or preventive trimming), pain, or lesion requiring treatment (LRT; hoof lesions treated with blocks or topical treatments) from the first trimming after enrollment are summarized in Table 1. These categories were used to generate 2 outcomes for this study, which were originally intended to be assigned based on lesions recorded at the follow-up evaluation. However, in cases where cows were trimmed earlier by the site hoof trimmer, the first postenrollment trimming record was used instead. The first outcome grouped cows as typically not treated (TNT), which included LNRT and pain, or typically treated (TT) representing cows with one or more LRT. The second outcome classified cows as typically treated or painful (TT+P), including those with at least one LRT or pain, or typically not treated and not painful (TNT-NP), which included LNRT. These outcome definitions served as the target labels for training and evaluating the machine learning classification models.

Description of Feature and Feature Sets. All features used in the analysis are summarized in Supplemental Table S1 (see Notes). Briefly, the features included both static and time-dependent components. Static features encompassed site, lactation number, DIM at follow-up, and condensed counts of health events occurring in the current lactation before the day of the outcome. The health

events included binary or categorical features indicating whether a cow experienced one or more events related to digestion, mastitis, metritis, transition, other health issues, or reproduction (categorized as 0–2 or 3+ events). Three additional binary features captured whether the cow had a prior history of infectious, noninfectious, or other hoof lesions.

Additionally, BCS, LS, and milk yield were treated as time-dependent features. They were calculated across 6 predefined time windows: 14, 28, 42, 56, 70, and 84 d before d 1, where d 1 was defined as the day immediately preceding the follow-up or hoof trimming evaluation. The decision to use biweekly (14-d) intervals was motivated by the high proportion of missing data in weekly BCS and LS measurements, with biweekly grouping substantially reducing the number of missing observations (Supplemental Table S2, see Notes). If more than one score existed for the day, then the earliest score was used.

For each time window, only data occurring within the defined time frame were used to generate features, regardless of whether additional data existed outside that window. Within each time window, features derived from LS, BCS, and milk yield included the average, maximum, and minimum, resembling those from Siachos et al. (2025), along with the creation of the median and range. For LS only, an additional feature was created that recorded the percentage of observations above 50 for each time window (company criteria for lameness). To be eligible for inclusion in each time window, cows were required to remain in the study for the full duration of the window without being trimmed during that period.

To evaluate different modeling scenarios, these features were organized into 16 feature sets. These sets were organized into 4 main groups: (1) all features, which included AUTO-derived features, site, and lactation; (2) no AUTO features, which excluded BCS and LS but could include site and lactation; (3) AUTO-only features, consisting of BCS and LS, with optional inclusion of site and lactation; and (4) LS-only features. Each group included both a base version and for elastic net a version that incorporated all possible 2-way interactions. When combined with the 6 different time windows used in modeling, this resulted in 96 total feature set and time window combinations.

Model Tuning, Training, and Testing. Once the data-sets were finalized, they were split into an 80% training set used for model development (training and tuning) and a 20% testing set reserved strictly for final model evaluation. Five machine learning algorithms were trained across both outcome definitions, feature sets, and time windows: elastic net, random forest, regularized discriminant analysis, gradient boosting, and k-nearest

neighbors (KNN). Only the elastic net models explicitly used interaction terms, as elastic net is a generalized linear model that requires interactions to be manually specified.

Except for KNN, each model underwent a 2-step tuning process to optimize performance, as summarized in Supplemental Table S3 (see Notes). In the first step, algorithm-specific hyperparameters were tuned using 10-fold cross-validation to minimize misclassification error. Data were processed within each fold to avoid data leakage. For elastic net and KNN, the training data within each fold was scaled using Z-score normalization, and the corresponding validation data were scaled using the mean and standard deviation derived from the training fold. For regularized discriminant analysis, the training data were scaled once before tuning, as the package used does not support per-fold standardization (Roever et al., 2023). For each model, feature set, and time window combination, the set of hyperparameters that achieved the lowest misclassification error in step 1 was carried forward for further tuning in the second step.

In the second step, a range of classification thresholds was evaluated using 10-fold cross-validation to determine the cutoff that maximized the $F_{0.5}$ score. During this process, elastic net and regularized discriminant analysis both used fold-wise scaling: training data within each fold were standardized, and the corresponding validation data were transformed using the training fold's parameters. This 2-step tuning process was repeated for every combination of outcome, feature set, and time window.

For each outcome, the single best-performing combination of feature set, time window, and model, determined by the highest $F_{0.5}$ score, were selected for final evaluation. In cases of ties within the same feature set, time window, and model combination, the configuration with the highest classification threshold was chosen. As a result, one model was selected for the TT+P versus TNT-NP outcome and one model for the TNT versus TT outcome. These top configurations were retrained using the full 80% training set and their respective hyperparameters. Data preprocessing steps, including any scaling procedures, were applied consistently with the approach used during step 2 for that specific model. Final model performance was assessed on the 20% testing set. Performance metrics included $F_{0.5}$, accuracy, PPV, negative predictive value, sensitivity, and specificity. Additionally, 95% CI were calculated for the training and testing performances only, using 10,000 bootstrap resamples with replacement. No CI were computed for the tuning performance, as model selection was based directly on the observed $F_{0.5}$ score during that phase.

Phase 2: Live Implementation Testing

Study Design and Housing. As a follow-up to phase 1, a live implementation of the top-performing model (random forest) was conducted from May to June 2025 at the same 2 dairy sites described earlier. The primary objective was to externally validate the model's performance on positive predictions, and the secondary objective was to compare the model's performance against the operation's current protocol (passive surveillance, defined as the routine, nonsystematic identification of cows for hoof trimming during daily farm activities without the use of structured locomotion scoring) for identifying cows in need of hoof trimming. To avoid influencing routine management practices, study personnel did not inform the operation staff responsible for passive surveillance about the trial. This project was approved under the University of Minnesota Institutional Animal Care and Use Committee protocol (2108-39336A).

Because the primary objective was to externally validate the PPV, sample size calculations followed the guidance of Whittle et al. (2025) for evaluating the external validation of the PPV of classification models. Based on results from phase 1, we assumed a sensitivity and PPV of 92.3% and a combined prevalence of 12.9%. A CI width of 0.3 was selected to reflect the preliminary nature of this study, which aims to evaluate model performance before committing resources to a full-scale assessment. This resulted in a sample size of 95 positive classifications at the operation level.

Eligibility Criteria. To create a list of cows eligible for external validation prediction the following criteria were used: eligible cows had to be lactating Holstein cows over 14 DIM producing saleable milk (not in the hospital pen), not on the trim list for a recheck, not been trimmed for in the last 90 d, not have more than 1 routine trimming or lesion recorded this current lactation, and no diagnosis of upper leg lameness in past or current lactations. These criteria were like those used for enrollment in phase 1.

Model Deployment. The random forest model from phase 1 was trained to classify cows as TT or TNT was applied at each site. The model used 14 d of pre-evaluation data and relied on the AUTO-only features.

To implement the model, cow-level data, including cow identification number and lactation number, were extracted from the farm's herd management software and downloaded as .csv files, containing only cows that met the eligibility criteria. Unlike in phase 1, AUTO data for phase 2 were retrieved directly from the CattleEye user platform (<https://cattleeye.com>; CattleEye Inc., Belfast, UK) and were generated using the same algorithm and camera setup described previously.

A data pipeline was developed by study personnel (D.S.) to automate the workflow. Each day before hoof trimming, study personnel (G.C.) uploaded files containing cow-level and AUTO data as described previously. The only manual input was the entry of the hoof trimming date (year, month, and day) and the hours of AUTO recording. Once this information was provided, the pipeline automatically executed all preprocessing steps, including feature generation and missing data checks. To ensure complete feature generation, eligible cows were excluded if they did not have at least one LS and one BCS observation within the full 14-d window preceding the evaluation.

Following preprocessing, the pipeline generated predictions and created hoof trimming lists. Only cows classified as TT were included, along with their predicted probability of having LRT. When trimming capacity was limited, cows were prioritized by their predicted probability, with the highest probable individuals selected first.

To resemble on-farm implementation, the regular hoof trimmers at each site conducted the hoof trimming assessments as part of their normal routine. Study personnel were responsible only for generating the trimming lists based on the model's predictions and available trimming slots. They did not assist in hoof trimming or lesion diagnosis.

Hoof Trimming and Farm Staff Data. Hoof trimming records were extracted from the operation's herd management software and downloaded as .csv files. Additionally, files containing cow ID, lactation, date, and flagged by farm staff were downloaded.

Data Management and Analysis

Missing Data. The proportions of cows ineligible for prediction due to missing LS, BCS, or both, were calculated separately and reported as crude percentages.

Evaluation. The primary objective of the live implementation was to evaluate the model's ability to correctly identify cows with hoof lesions among those it flagged as positive. Positive predictive value was the main performance metric assessed and was calculated as the proportion of flagged and trimmed cows confirmed to have an outcome reported by the hoof trimmer. The PPV were calculated separately for any outcome detected by the hoof trimmer (any outcome) and for the predefined lesions of interest (original outcome: M2, sole ulcer, toe lesion, white line disease, or other), for both overall and by site. The same PPV calculations were performed for passive surveillance, where a cow was considered flagged if it was recorded in the herd management staff as lame within 14 d before the trimming date. For both the random forest model and passive surveillance, 95% CI for PPV

were calculated using bootstrap resampling, performed separately for each site as described for phase 1.

RESULTS

Phase 1: Model Development

Descriptive Data. As reported in Swartz and Cramer (2026), 522 cows were initially assessed at enrollment. Of these, 2% (11/522) were ineligible to continue and 10% (54/522) were lost to follow-up, primarily due to removal from the herd (Figure 1). In total, 88% (457/522) of cows had follow-up data available. Enrollment demographics by site are presented in Table 2.

Overall, most cows were classified as LNRT (TNT-NP) at follow-up (Figure 2), with heel erosion, sole hemorrhage, and white line hemorrhage being the most frequent lesions observed (Supplemental Figure S1, see Notes). For the TT+P outcome, pain was the most frequently observed outcome, followed by white line disease and M2 lesions (Figure 3). Although the incidence rates differed slightly between sites, the relative frequency of these outcomes remained consistent. It is important to note that depending on the dataset and time interval, some cows were excluded from feature sets due to either insufficient time between enrollment and follow-up evaluation or missing data (Figure 4). The specific composition of the training and testing sets used for training and testing is presented in Supplemental Table S4 (see Notes) for the TNT-NP versus TT+P outcome and in Supplemental Table S5 (see Notes) for the TNT versus TT outcome.

Model Training and Testing Performance. The overall performance of all models after phase 2 of tuning is available in the supplemental material (see Notes). The highest-performing model, ranked by their $F_{0.5}$ scores, along with their tuning performance and corresponding hyperparameters for each outcome, are summarized in Table 3.

The highest-performing model was the random forest for the TNT versus TT outcome, which used 14 d of BCS and LS based features. For the TNT-NP versus TT+P outcome, the best model was the regularized discriminant analysis using 42 d of BCS- and LS-based data. During the tuning phase, $F_{0.5}$ scores were generally low due to poor sensitivity across both outcomes. When these models were retrained on their respective full training set and evaluated on the same data (i.e., training performance) and testing set (i.e., testing performance), $F_{0.5}$ scores improved substantially for the random forest model, and decreased slightly for the regularized discriminant analysis model (Table 3). When evaluating feature importance for the retrained random forest model, LS features demonstrated greater importance than BCS features (Figure 5). Specifically, average and median LS resulted in the

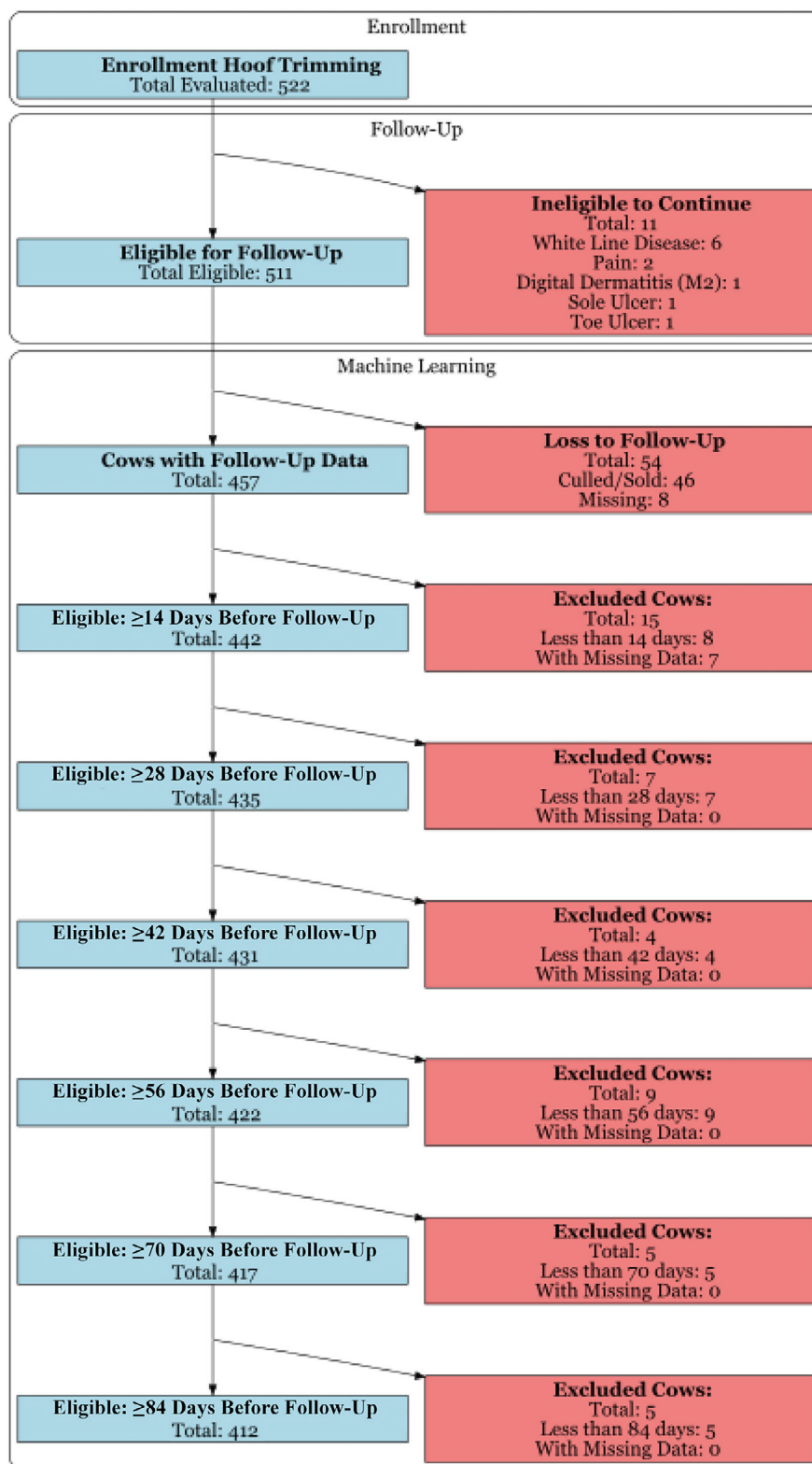


Figure 1. Flowchart illustrating the progression of cows through the study. The chart begins with cows that underwent enrollment hoof trimming, followed by those deemed ineligible to continue. Of the cows eligible for follow-up, a subset was lost to follow-up. Among the cows with follow-up data, additional subsets were excluded at each time point due to missing data or insufficient days between enrollment and follow-up. The remaining cows with valid data at each time point (e.g., 14, 28, 42, 56, 70, and 84 d before outcome evaluation) are shown sequentially. The final group consists of cows with complete data for all 84 d.

Table 2. Demographic characteristics of cows that remained in the study from enrollment through follow-up by site (number, with percentage in parentheses); variables include DIM group at enrollment, lactation number, and hoof lesion history

Item	Site 1 (n = 228)	Site 2 (n = 229)	Overall (n = 457)
DIM at enrollment			
0–30	70 (30.7)	36 (15.7)	106 (23.2)
31–60	81 (35.5)	91 (39.7)	172 (37.6)
61–100	77 (33.8)	102 (44.5)	179 (39.2)
Lactation			
1	0 (0)	229 (100)	229 (50.1)
2	120 (52.6)	0 (0)	120 (26.3)
3	58 (25.4)	0 (0)	58 (12.7)
4	34 (14.9)	0 (0)	34 (7.4)
5	12 (5.3)	0 (0)	12 (2.6)
6	4 (1.8)	0 (0)	4 (0.9)
Infectious lesion history ¹			
0	194 (85.1)	221 (96.5)	415 (90.8)
1	32 (14.0)	8 (3.5)	40 (8.8)
2	1 (0.4)	0 (0)	1 (0.2)
3	1 (0.4)	0 (0)	1 (0.2)
Noninfectious lesion history ²			
0	211 (92.5)	229 (100)	440 (96.3)
1	17 (7.5)	0 (0)	17 (3.7)
Other hoof lesion history ³			
0	223 (97.8)	228 (99.6)	451 (98.7)
1	5 (2.2)	1 (0.4)	6 (1.3)

¹The count of prior infectious hoof lesions, specifically digital dermatitis, foot rot, or treatments with oxytetracycline or Excede (Zoetis). This includes events from the current and previous lactation only.

²The count of prior noninfectious hoof conditions, specifically white line disease, toe ulcers, and sole ulcers. This includes events from the current and previous lactation only.

³Count of different lesions reported by the site hoof trimmers in the current and previous lactation: interdigital hyperplasia, periople ulcer, non-ulcer, other, puncture, vertical fissure, axial fissure, corkscrew.

largest mean decrease in accuracy, whereas average and maximum LS contributed the most to the mean decrease in Gini (the average decrease in node impurity across all trees in the forest). However, average BCS ranked third and fourth in mean decrease in accuracy and Gini, respectively.

Phase 2: Live Implementation and Generalizability Testing

Descriptive Data. Across the 2 sites, 15 hoof trimming lists were developed, resulting in a total of 6,304 records that met the eligibility criteria and were extracted from the dairy herd management software. Of these, 3% (180/6,304) were not eligible for random forest model prediction due to missing data. The majority of ineligible observations occurred at site 2 and were primarily due to missing both LS and BCS (Figure 6). As a result, 97.1% (6,124/6,304) of cows had sufficient AUTO data to be eligible for prediction.

Validation Performance. Among all records, 2.6% (161/6,304) were classified as positive by the random forest model, compared with 1.1% (70/6304) flagged by passive surveillance. Random forest had 86.1% (161/187) positively classified cows hoof trimmed. The

PPV was 25.5% (41/187; 95% CI: 21.9–30) for any outcome and 13.7% (22/187; 95% CI: 9.9%–17%) for TT outcomes. In comparison, cows flagged by passive surveillance had 100% (70/70) positively hoof trimmed, with a PPV of 67.1% (47/70; 95% CI: 63.2%–72.6%) for any outcome and 48.6% (34/70; 95% CI: 40%–57.9%) for TT outcomes. Thus, passive surveillance thus yielded a higher PPV than random forest for both outcomes of interest (Figure 7).

Across both detection approaches, 78 cows were identified with an outcome at hoof trimming. Of these, 12.8% (10/78) were flagged by both approaches, 39.7% (31/78) were identified by the random forest model only, and 47.4% (37/78) by passive surveillance alone (Figure 8).

DISCUSSION

During the first phase of this study, the primary objective was to develop and tune machine learning algorithms for classifying lactating dairy cows with hoof lesions, with a focus on maximizing the $F_{0.5}$ score. Among all models evaluated, a random forest model trained on 14 d of data using AUTO-derived features outperformed all alternatives for the classification of TNT versus TT during the tuning phase. During the second phase of this study,

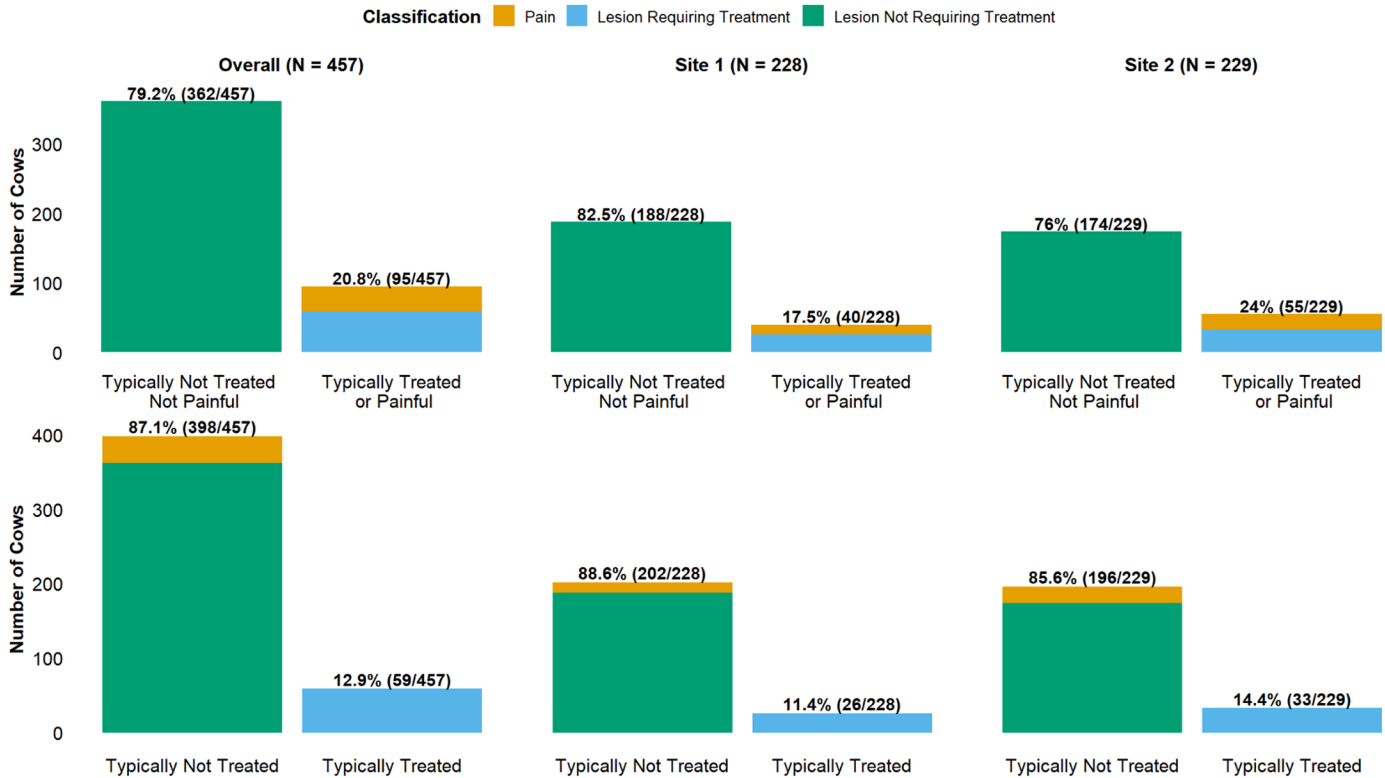


Figure 2. Distribution of outcomes by site under 2 classification approaches: typically not treated and not painful versus typically treated or painful (top) and typically not treated versus typically treated (bottom). Bars represent the number of cows within each outcome category, colored by outcome types. Labels above each bar indicate the percentage and proportion of cows in each outcome by site.

the primary objective was to perform live implementation of the random forest model and to evaluate the PPV and compare its performance to that of the operation's current protocol. Overall, the PPV of the random forest model during live implementation was lower compared with its testing performance and passive surveillance. Although passive surveillance was more precise, each method identified different cows. The model may help to detect cows missed by farm staff, but unnecessary trims could increase costs or limit trimming capacity.

Our modeling approach prioritized PPV over sensitivity to support real-world adoption, where it is a key metric for producer confidence and value (Van Nuffel et al., 2015; O'Leary et al., 2020). This required accepting a trade-off in identifying all cows with hoof lesions, which was achieved with the random forest model for the TNT versus TT outcome. However, because cows were reclassified on a weekly basis, we viewed this as an acceptable compromise, offering more detection opportunities than current practices. For example, Logan et al. (2024) reported that locomotion scoring has poor sensitivity and a low to moderate PPV, despite being intentionally conducted. In contrast, most farms rely on passive sur-

veillance, which has been associated with increased odds of lameness compared with regular locomotion scoring, likely due to delays in detecting and treating affected animals (Groeneveld et al., 2014). Because producers tend to identify only severely lame cows as lame (Leach et al., 2010), passive surveillance may yield a relatively high positive predictive value, as our results indicated, but this comes at the cost of missing earlier cases. Furthermore, passive surveillance results in few cows being observed daily, and detection relies on incidental observation. In comparison, our approach attempted to evaluate each cow at minimum weekly, addressing the limitations of passive surveillance. However, missing AUTO data reduced the consistency of weekly evaluations and limited the model's ability to fully address the gaps in passive surveillance.

Additionally, an algorithm with high sensitivity and low PPV for treatable lesions would detect more cows needing treatment but would also increase the frequency of unnecessary trimming. Frequent flagging and hoof trimming of nonlesion cows increases the risk of over-trimming and thin soles (Cramer and Solano, 2024). Although introducing a waiting period between trims could

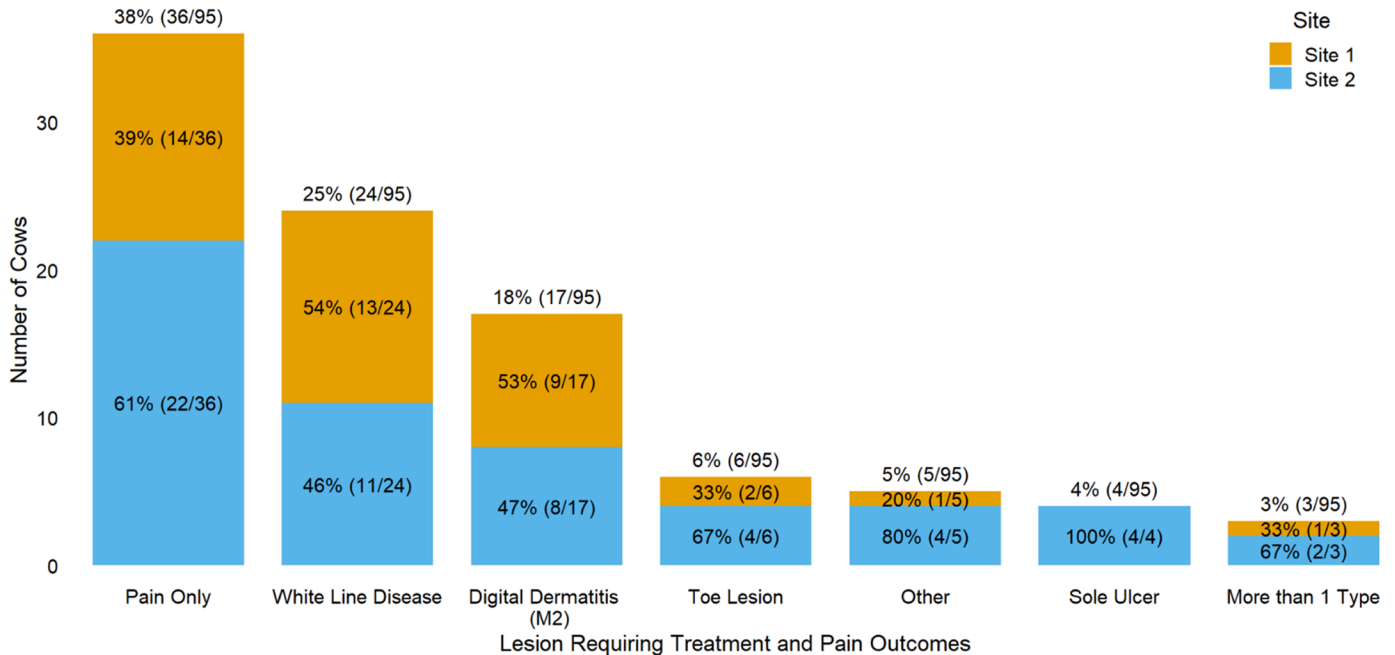


Figure 3. Distribution of lesion requiring treatment and pain outcome types diagnosed at follow-up, stacked by site. Bars represent the number of cows diagnosed with each lesion type at follow-up evaluation. Bars are stacked by site (site 1 and site 2), and labels within the bars indicate the percentage and proportion contribution of each site to the outcome type. The label above each bar denotes the total percentage and proportion of cows diagnosed with that outcome type across both sites. The “other” category includes lesions reported by site hoof trimmers that consists of: interdigital hyperplasia, periople ulcer, non-ulcer, other, puncture, vertical fissure, axial fissure, and corkscrew claw. The “more than 1 type” category includes cows diagnosed with multiple lesion types at follow-up: at site 1, this included white line disease and M2; at site 2, one cow had both white line disease and a toe ulcer, and another had both a sole ulcer and M2.

mitigate this, it also raises the risk of missing lesion development during the interim. To balance detection with trimming precision, we used the $F_{0.5}$ score, which emphasizes PPV while still incorporating sensitivity, identifying more true cases without substantially increasing false positives. An algorithm optimized in this way, when paired with more strategic routine trimming, could reduce unnecessary hoof trimmings, labor costs, and short-term impacts such as reduced milk yield (Stoddard and Cramer 2024).

To ensure relevance for on-farm decision-making, we selected outcome labels that are actionable and aligned with treatment practices. However, classification was complicated by cows that exhibited pain. Because LS is an indicator of impaired locomotion, often a result of pain (Cramer et al., 2025), we evaluated pain on all hooves at the follow-up evaluation using a hoof tester to elicit a withdrawal reflex. Hoof trimmers can use hoof testers to assess for a withdrawal reflex to detect painful hoof lesions, although this is not universally practiced. The presence of a withdrawal reflex without a visible lesion raises questions about its clinical significance. It may represent the early onset of a lesion not yet visible, or it

could reflect transient or unrelated discomfort that may not progress. To account for this ambiguity, we evaluated 2 outcome definitions: one combining pain with LNRT (TNT vs. TT) and another combining LRT with pain (TNT-NP vs. TT+P). This approach allowed us to assess model performance under both assumptions and reflect the ambiguity present in field conditions.

Across the 2 outcome definitions evaluated, the highest $F_{0.5}$ score on the testing set was achieved by the TNT versus TT classification. This model was a random forest, in which cows classified as pain were grouped with LNRT. To understand why the random forest model was successful, it is important to understand how it operates. As described by Sarker (2021), random forest is an ensemble method that constructs multiple decision trees using randomly selected subsets of data and features, combining their outputs to improve prediction accuracy and reduce overfitting. In contrast, for the TNT-NP versus TT+P outcome, regularized discriminant analysis performed best during the tuning phase but failed to improve in the testing phase. This drop in performance may be due to the model’s reliance on linear relationships (Friedman, 1989), which may not have held consistently across the

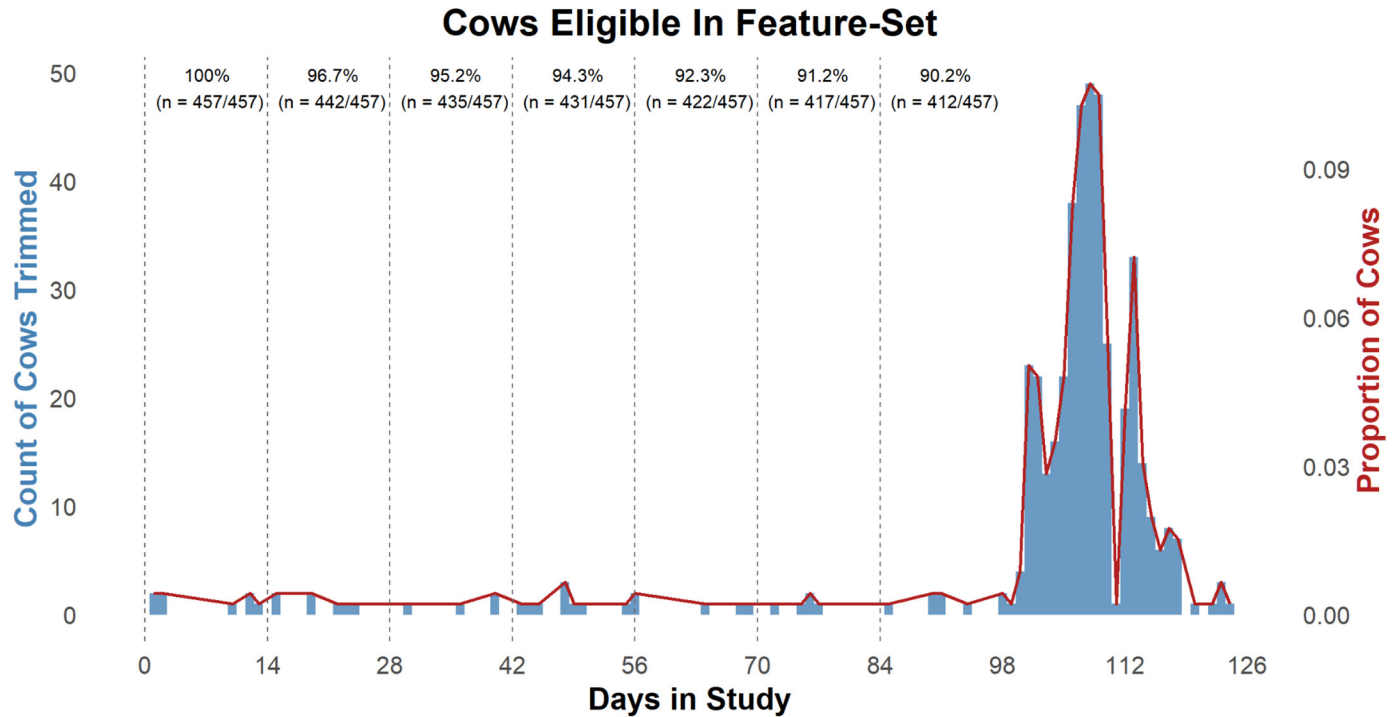


Figure 4. Epicurve showing the number and proportion of cows trimmed by days in study. The blue area represents the count of cows that underwent follow-up evaluation (hoof trimming) on each day in the study, and the red line indicates the proportion of cows trimmed relative to the full study population. Vertical dashed lines represent the time windows used for feature construction (14, 28, 42, 56, 70, and 84 d before follow-up). At each time point, the percentage and number of cows remaining eligible for analysis are displayed above the plot. Cows were excluded at each window due to missing data or insufficient time between enrollment and follow-up.

testing data. Alternatively, the poor performance of the regularized discriminant analysis may also suggest that grouping pain with LRT was not the optimal outcome definition. However, this remains uncertain, as the clinical relevance of pain in the absence of visible lesions is not yet well understood. Further research is needed to determine whether cows exhibiting a withdrawal reflex without an observable hoof lesion are likely to develop a lesion over time if no intervention is provided.

Additionally, for TNT versus TT, the random forest model relied on only 14 d of data before the hoof trimming event and included only BCS and LS from AUTO. Although LS features demonstrated greater overall importance, BCS features, particularly average BCS, also contributed meaningfully. Although the specific thresholds used within each decision tree to split BCS and LS features are not available, the reported relationships between these variables and hoof lesions may offer insight. For example, Whay et al. (1998) demonstrated that different lesion types can have varying effects on pain expression and, consequently, on locomotion. In addition, low BCS have been associated with a higher probability of developing lameness (Lim et al., 2015) and hoof lesions (Green et al., 2014). These associations

may help to explain why the model incorporating both BCS and LS outperformed the model using LS alone.

The strong model performance using only the 14-d window before the follow-up evaluation suggests that the clinical signs began during this period. This date range is similar to previous studies that have looked at the temporal association between LS and the diagnosis of hoof lesions. For example, Swartz et al. (2024) described that cows without lesions had slightly lower score distribution than those with lesions as early as 4 wk before diagnosis, with the largest increase between wk 2 and 1 before trimming. Similarly, Siachos et al. (2025) found that cows with severe lesions had higher scores than those with moderate or mild lesions beginning 23 d before trimming. However, both studies defined lesion diagnosis relative to diagnosis, which may not reflect actual onset.

Although a 7-d window may have improved performance, it was not feasible due to a high percentage of missing weekly observations. In our study, reduced model performance when including data beyond 14 d suggests that BCS and LS were more similar between TNT and TT cows in earlier time windows. As a result, including earlier observations may have diluted the impact of changes

Table 3. Summary of the top-performing models¹ selected for each outcome: lesion not requiring treatment (LNRT) with or without pain, grouped as typically not treated (TNT), versus lesion requiring treatment (LRT), classified as typically treated (TT), and LNRT, grouped as typically not treated and not painful (TNT-NP), versus lesion requiring treatment (LRT) with or without pain, classified as typically treated or painful (TT+P), based on the $F_{0.5}$ score during the tuning phase

Outcome	Model ³	Days ⁴	Version ⁵	Hyperparameters ⁶	Phase	Performance metrics ²					
						$F_{0.5}$	ACC	SENS	SPEC	PPV	NPV
TNT-NP vs. TT+P	RDA	42	AUTO	Gamma = 0.541 Lambda = 0.643 Threshold = 0.49	Tuning	47.6	81.7	26.1	95.6	60	83.8
					Training	48.7	82	26.1	96	62.1	83.8
					Testing	(33.7, 61.8)	(77.9, 86)	(16.2, 36.7)	(93.5, 98.2)	(44, 80)	(79.6, 87.8)
TNT vs. TT	RF	14	AUTO	ntree = 600 mtry = 3 nodesize = 1 threshold = 0.5	Testing	43.4	80.5	22.2	95.7	57.1	82.5
					Tuning	(14.8, 69.4)	(71.3, 88.5)	(5, 44)	(90.3, 100)	(16.7, 100)	(73.8, 90.2)
					Training	59.1	90.7	31	98.7	76.5	91.4
					Testing	100	100	100	100	100	100
						(100, 100)	(100, 100)	(100, 100)	(100, 100)	(100, 100)	(100, 100)
						92.3	97.8	92.3	98.7	92.3	98.7
						(76.4, 100)	(94.4, 100)	(75, 100)	(95.8, 100)	(75, 100)	(95.8, 100)

¹For each model, the outcome, algorithm, number of days of data used, feature set version, and selected hyperparameters are listed.

²Performance metrics are reported across 3 phases, tuning, training, and testing, and include $F_{0.5}$, accuracy (ACC), sensitivity (SENS), specificity (SPEC), positive predictive value (PPV), and negative predictive value (NPV). Metrics from the training and testing phases reflect performance on the respective datasets, reported as estimate (95% CI), with CI generated from 10,000 bootstrap samples.

³RDA = regularized discriminant analysis; RF = random forest.

⁴Number of days before the outcome evaluation that is included in the data.

⁵AUTO only features hyperparameters identified during the tuning phase.

⁶The hyperparameters associated with each model.

closer to lesion development. These findings suggest that a 2-wk window is sufficient for identifying cows with hoof lesions.

In addition to evaluating 2 outcomes, model selection was a result of using multiple machine learning algorithms and feature sets. The use of multiple feature sets and machine learning algorithms in this study should not be considered a methodological flaw or a type of “p-hacking” (Head et al., 2015). This study focused on prediction, not causal inference (Calvache et al., 2023), and all model development and hyperparameter tuning were conducted exclusively on the training set. The testing set remained untouched until final evaluation to prevent data leakage and produce unbiased model performance estimates (Hastie et al., 2009). Hyperparameter tuning was performed using 10-fold cross-validation, a standard method that balances bias and variance, producing accurate and stable performance estimates while minimizing overfitting (Hastie et al., 2009).

Although our model demonstrated strong internal performance, when we externally validated our model, we observed a decrease in PPV. This aligns with the conclusions of Bleeker et al. (2003), who noted that internal validation of prediction models using bootstrap methods for model performance may be insufficient for assessing future performance, particularly when using small datasets such as the phase 1 testing set. One possible reason for the decrease in performance could be the result of overfitting of the original training data. Although both the training and real-world application took place at the same sites, they involved different cohorts. The live implementation involved a new cohort of cows, whose characteristics may have differed from the training population (J. Gu, University of Munich, Munich, Germany, and Siemens AG, Corporate Technology, Munich, Germany; D. Oelke, Siemens AG, Corporate Technology, Munich, Germany; unpublished data). Ultimately, this decline reinforces the concern that strong internal performance does not guarantee external validity (Swartz, 2025).

In dairy production, this risk of overfitting and poor generalization is amplified due to variation in herd management, environmental conditions, and population characteristics across farms. Our dataset from phase 1 reflects several of these limitations. The training set was composed predominantly of TNT cows, and a disproportionate number of observations came from site 2 for the testing set, which consisted exclusively of first-lactation cows. Furthermore, because the AUTO system has not been validated on other breeds, model development only included Holsteins. These factors may affect the model’s ability to generalize to other dairy systems. Given that most machine learning studies classifying lameness or hoof lesions do not report performance beyond their

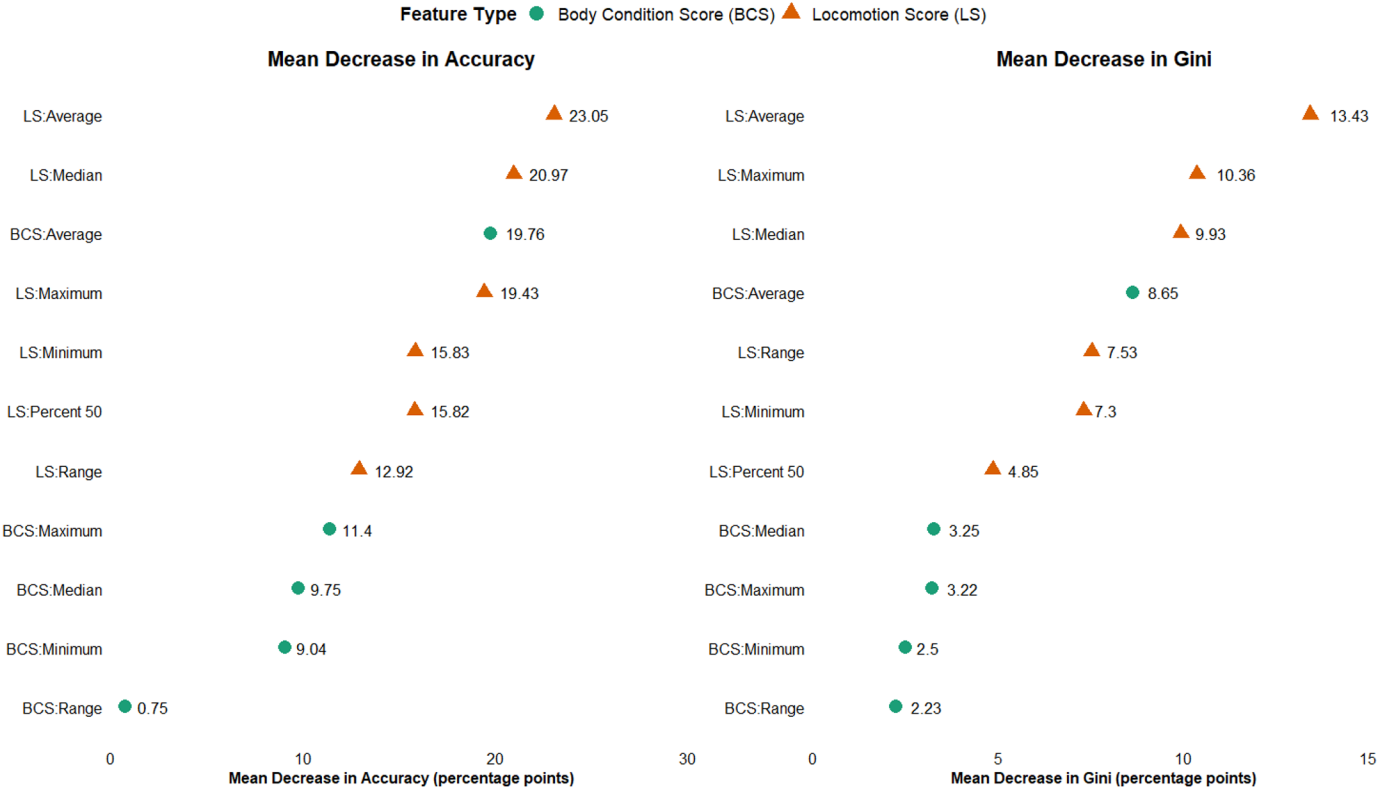


Figure 5. Variable importance plot from the retrained random forest model developed to classify cows as typically not treated versus typically treated, based on features collected during the 14 d before the hoof trimming evaluation. The plot displays the mean decrease in accuracy (the average reduction in model prediction accuracy when a given feature is removed) and the mean decrease in Gini (the average decrease in node impurity across all trees in the forest).

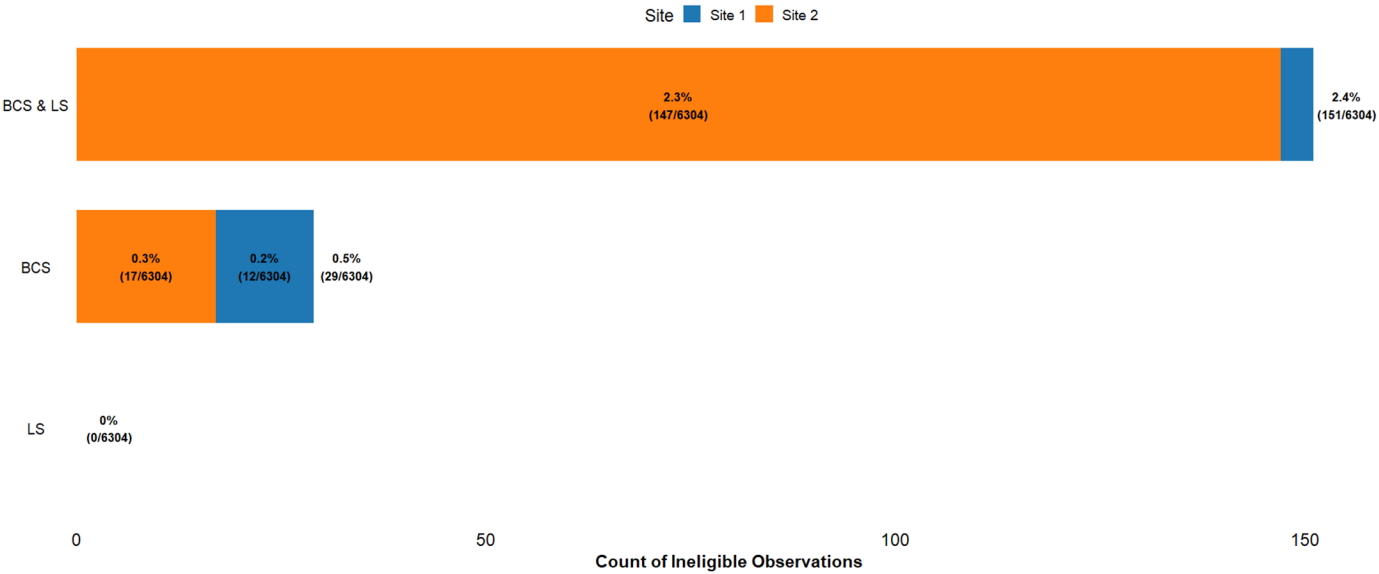


Figure 6. Percentage and proportion of observations ineligible for prediction due to missing data on locomotion scores, BCS, or both. Observations were classified as ineligible if either BCS or locomotion scores had no recorded values during the 14 d before the hoof trimming evaluation.

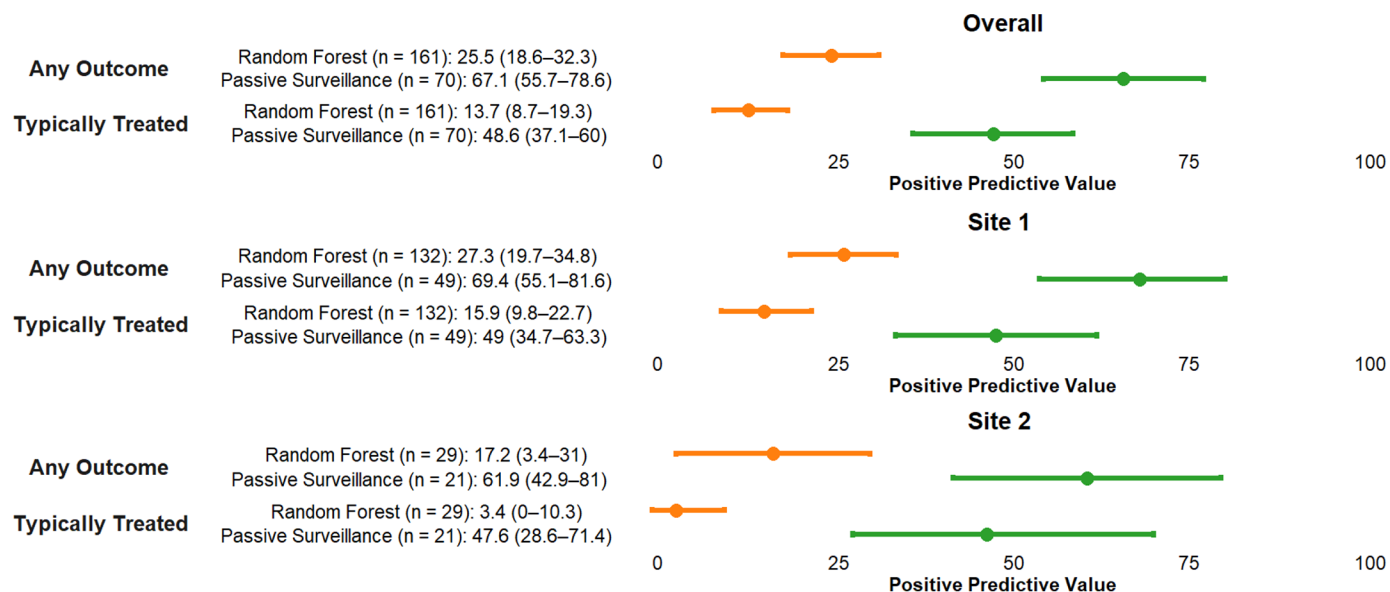


Figure 7. Predictive performance of the model applied in the field, summarized by site. Bars represent the positive predictive value for cows identified as positive by the model and sent to the hoof trimmer. “Any outcome” includes any nonhealthy follow-up diagnosis recorded by the hoof trimmer, and “original outcome” refers specifically to hoof lesions targeted by the model. Estimates are shown with 95% CI based on 10,000 bootstrap resamples with replacement.

training and testing datasets (Swartz, 2025), we prioritized real-time evaluation at the original sites before allocating additional resources for external evaluations.

Another factor that may have contributed to the reduced model performance during live implementation is variation in outcome recording. The hoof trimmers



Figure 8. Overlap between cows flagged for hoof trimming by the random forest model and passive surveillance (passive). The diagram shows the number of cows identified by the random forest model, by passive surveillance, and by both methods. Text within each section includes the number of cows identified with hoof lesions and a breakdown of lesion outcomes observed during trimming.

conducting the follow-up evaluations may not have examined the cows as thoroughly as the research personnel, contributing to the reduced detection of lesion outcomes. Although the hoof trimmers were trained by study personnel, their standardized lesion recording was not updated before the live implementation. Though we did not assess this directly, future on-farm applications should incorporate role-specific training to ensure consistent and accurate use of the technology. This helps ensure that reduced performance in practice is not due to inconsistent application or evaluation procedures but instead reflects true model limitations. Regardless of whether the decline in positive predictive value was due to population differences or inconsistent outcome recording, it reflects the realities of on-farm use and highlights the need for real-world validation before recommending a tool for adoption.

In addition to the decrease in PPV, passive surveillance demonstrated higher predictive performance and resulted in fewer unnecessary trims than the random forest model. However, only a small number of cows were identified by both methods, indicating limited overlap in the cows flagged. Passive surveillance likely identified cows with more obvious clinical signs, whereas the random forest model flagged cows that were not detected through routine observation, resulting in different subsets of animals being identified.

Given that producers often reserve the term “lameness” for severely lame cows (Leach et al., 2012), methods relying on observable gait abnormalities may under-recognize lesions that do not consistently manifest as overt lameness. This may help to explain why the random forest model identified more cows with digital dermatitis, whereas passive surveillance identified more cows with white line disease. Evidence from autonomous locomotion data supports this pattern, as Swartz et al. (2024) reported that digital dermatitis was characterized by low and relatively consistent LS over time, whereas white line disease exhibited the greatest increase in LS. As passive surveillance emphasizes obvious visible gait abnormalities, digital dermatitis may be under-detected, whereas white line disease may be more readily identified, although detection may depend on lesion severity. Taken together, these findings suggest that the random forest model may operate in parallel with existing farm protocols by identifying a different subset of cows than passive surveillance. Although hoof lesion or lameness severity data were not collected in this study, early identification of hoof lesions could present opportunities for intervention before the onset of lameness. Nonetheless, the model’s lower PPV may be viewed unfavorably by producers.

Although low PPV can lead to added expenses, it does not fully capture the system’s economic return. As noted in Swartz (2025), the cost-to-benefit ratio is one of the primary barriers to the adoption of lameness detection technologies. However, evaluating the potential savings from correctly identifying affected cows is complicated by the wide variability in lameness and hoof lesion cost estimates reported in the literature (Dolecheck and Bewley, 2018). These estimates depend on factors such as parity and stage of lactation (Dolecheck et al., 2019) and are based on comparisons between lame and nonlame cows, as well as cows with and without hoof lesions. To our knowledge, no study has directly assessed the economic impact of treating versus not treating hoof lesions. Nonetheless, delayed treatment is known to reduce recovery likelihood, with acutely lame cows showing significantly better recovery outcomes than chronically lame cows (Thomas et al., 2015, 2016). It remains unclear whether treating hoof lesions in cows that remain lame results in economic savings, although intervention may be warranted from a welfare perspective. As discussed in Swartz (2025), future work should evaluate the timeliness of detection and assess whether treatment improves both clinical and economic outcomes.

To determine the economic impact of a system with a low PPV, it is important to consider the impact of earlier detection on the cost of a first-time lameness event, which has been estimated at \$500 to \$900 (Puerto et al., 2021). This potential reduction in cost represents a benefit of implementing the system. However, a test with low PPV also incurs short-term costs associated with trimming and reductions in milk yield (Stoddard and Cramer, 2024). Although these costs may be justifiable if early intervention prevents lesion progression, the low PPV may lead to additional economic consequences when hoof trimming capacity is limited. In such cases, not all cows flagged by the model are able to be trimmed. When hoof trimming slots are limited, false positives become more costly, as they can displace cows that would benefit from treatment, potentially leading to worse outcomes and future financial loss. In this context, the misallocation of hoof trimming slots caused by false positives reduces the potential savings by screening more cows and inaccurately selecting cows for the hoof trimming. However, as passive surveillance and the random forest identified different cows and positive predictions were not always limited from hoof trimming capacity, there may be potential benefits for operations with underutilized hoof trimming slots. In larger herds with few free trimming slots, the margin of error may be smaller. Because this study focused on PPV and did not track all cows that were scored by the model and passive surveil-

lance, we were unable to identify how many cows were misallocated trim opportunities (sensitivity).

A key strength of this study was the use of a case-cohort design, as described by O'Brien et al. (2022). We identified a clearly defined population of cows in early to mid-lactation, confirmed they were free of hoof lesions and pain at enrollment, and followed them forward to observe outcome development. At the time of enrollment, all cows effectively served as their own controls. An additional strength was the real-world deployment of the model. Unlike models evaluated only in theoretical or retrospective settings, ours was tested in live production environments, providing practical insight into its strengths and limitations for on-farm decision-making. However, future studies should validate this performance on other dairy operations.

However, the design also introduced some limitations. Because we focused on mid-lactation cows with a maximum of 1 previous noninfectious hoof lesion, the model may not generalize to cows with a history of hoof lesions. As a result, cows prone to recurrent lesions may be underrepresented or misclassified. Furthermore, because the live implementation only assessed PPV, it does not provide insight into the model's sensitivity, specificity, or ability to rule out cows without lesions (i.e., negative predictive value) as the only cows evaluated were those with positive predictions.

After reflecting on our work, we propose that the relationship between locomotion scoring and hoof lesions reflects a potential paradigm shift in how hoof lesions should be detected. Although Logan et al. (2024) reported poor sensitivity of LS in detecting lesions, this highlights an important distinction that although most lame cows have hoof lesions, not all cows with hoof lesions present with lameness (van Huyssteen et al., 2020). This challenges the conventional reliance on lameness as an indicator of hoof lesion presence and supports a shift in focus toward identifying lesions earlier in their development before the onset of lameness. We recommend that future studies explore detection strategies that do not depend on traditional locomotion scoring but instead aim to identify earlier signs of lesion development before the onset of lameness.

CONCLUSIONS

Hoof lesion detection remains a significant challenge in lameness prevention and management. In this study, we developed and evaluated multiple machine learning approaches to detect cows with hoof lesions. Of the approaches evaluated, a random forest model incorporating BCS and scores from the AUTO technology achieved the highest $F_{0.5}$ score during development. Although the random forest model had a strong PPV internally, live

implementation revealed a lower PPV than the operation's current approach (passive surveillance). Notably, the 2 methods identified different cows. Based on these findings, we do not support the current adoption of our algorithm based on AUTO scores for hoof lesion detection. Further refinement and external validation or a financial benefit obtained across more herds will be necessary before this tool can be recommended for widespread on-farm use.

NOTES

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Nonstandard abbreviations used: ACC = accuracy; AUTO = autonomous camera system; KNN = k-nearest neighbors; LNRT = lesion not requiring treatment; LRT = lesion requiring treatment; LS = locomotion score; M2 = digital dermatitis ulcerative lesion ≥ 2 cm in diameter; NA = not applicable; NPV = negative predictive value; P1 = first phalanx; P2 = second phalanx; PPV = positive

predictive value; SENS = sensitivity; SPEC = specificity; TNT = typically not treated; TNT-NP = typically not treated and not painful; TT = typically treated; TT+P = typically treated or painful.

REFERENCES

- Adams, A. E., J. E. Lombard, C. P. Fossler, I. N. Román-Muñiz, and C. A. Kopral. 2017. Associations between housing and management practices and the prevalence of lameness, hock lesions, and thin cows on US dairy operations. *J. Dairy Sci.* 100:2119–2136. <https://doi.org/10.3168/jds.2016-11517>.
- Anagnostopoulos, A., B. E. Griffiths, N. Siachos, J. Neary, R. F. Smith, and G. Oikonomou. 2023. Initial validation of an intelligent video surveillance system for automatic detection of dairy cattle lameness. *Front. Vet. Sci.* 10:1111057. <https://doi.org/10.3389/fvets.2023.1111057>.
- Archer, S., N. Bell, and J. Huxley. 2010. Lameness in UK dairy cows: A review of the current status. *In Pract.* 32:492–504. <https://doi.org/10.1136/inp.c6672>.
- Berry, S. L., D. H. Read, R. L. Walker, and T. R. Famula. 2010. Clinical, histologic, and bacteriologic findings in dairy cows with digital dermatitis (footwarts) one month after topical treatment with lincomycin hydrochloride or oxytetracycline hydrochloride. *J. Am. Vet. Med. Assoc.* 237:555–560. <https://doi.org/10.2460/javma.237.5.555>.
- Bleeker, S. E., H. A. Moll, E. W. Steyerberg, A. R. T. Donders, G. Derksen-Lubsen, D. E. Grobbee, and K. G. M. Moons. 2003. External validation is necessary in prediction research: A clinical example. *J. Clin. Epidemiol.* 56:826–832. [https://doi.org/10.1016/S0895-4356\(03\)00207-5](https://doi.org/10.1016/S0895-4356(03)00207-5).
- Calvache, J. A., C. Higgins Tejera, J. A. Calvache, and C. H. Tejera. 2023. The three main tasks of modern epidemiology: description, prediction and causal inference. *Colombian Journal of Anesthesiology* 51:e1088. <https://doi.org/10.5554/22562087.e1088>.
- Cha, E., J. A. Hertl, D. Bar, and Y. T. Gröhn. 2010. The cost of different types of lameness in dairy cows calculated by dynamic programming. *Prev. Vet. Med.* 97:1–8. <https://doi.org/10.1016/j.prevetmed.2010.07.011>.
- Cook, N. B., J. P. Hess, M. R. Foy, T. B. Bennett, and R. L. Brotzman. 2016. Management characteristics, lameness, and body injuries of dairy cattle housed in high-performance dairy herds in Wisconsin. *J. Dairy Sci.* 99:5879–5891. <https://doi.org/10.3168/jds.2016-10956>.
- Cramer, G., K. D. Lissemore, C. L. Guard, K. E. Leslie, and D. F. Kelton. 2008. Herd- and cow-level prevalence of foot lesions in Ontario dairy cattle. *J. Dairy Sci.* 91:3888–3895. <https://doi.org/10.3168/jds.2008-1135>.
- Cramer, G., and L. Solano. 2024. Lameness Originating in the Hoof in Cattle. Accessed Jan. 13, 2025. <https://www.merckvetmanual.com/musculoskeletal-system/lameness-in-cattle/lameness-originating-in-the-hoof-in-cattle>.
- Cramer, G., L. Solano, and P. Constable. 2025. Detection and Diagnosis of Lameness in Cattle. Accessed Jun. 12, 2025. <https://www.merckvetmanual.com/musculoskeletal-system/lameness-in-cattle/detection-and-diagnosis-of-lameness-in-cattle>.
- DeFrain, J. M., M. T. Socha, and D. J. Tomlinson. 2013. Analysis of foot health records from 17 confinement dairies. *J. Dairy Sci.* 96:7329–7339. <https://doi.org/10.3168/jds.2012-6017>.
- Dolecheck, K., and J. Bewley. 2018. Animal board invited review: Dairy cow lameness expenditures, losses and total cost. *Animal* 12:1462–1474. <https://doi.org/10.1017/S1751731118000575>.
- Dolecheck, K. A., M. W. Overton, T. B. Mark, and J. M. Bewley. 2019. Use of a stochastic simulation model to estimate the cost per case of digital dermatitis, sole ulcer, and white line disease by parity group and incidence timing. *J. Dairy Sci.* 102:715–730. <https://doi.org/10.3168/jds.2018-14901>.
- Egger-Danner, C., P. Nielsen, A. Fiedler, A. Müller, T. Fjeldaas, D. Döpfer, V. Daniel, C. Bergsten, G. Cramer, A. M. Christen, K. F. Stock, G. Thomas, M. Holzhauser, A. Steiner, J. Clarke, N. Capion, N. Charfeddine, J. Pryce, E. Oakes, J. Burgstaller, B. Heringstad, C. Ødegård, J. Kofler, F. Egger, and J. B. Cole. 2020. ICAR Claw Health Atlas. 2nd ed. ICAR Technical Series. International Committee for Animal Recording, Rome, Italy.
- Ferguson, J. D., D. T. Galligan, and N. Thomsen. 1994. Principal descriptors of body condition score in Holstein cows. *J. Dairy Sci.* 77:2695–2703. [https://doi.org/10.3168/jds.S0022-0302\(94\)77212-X](https://doi.org/10.3168/jds.S0022-0302(94)77212-X).
- Friedman, J. H. 1989. Regularized Discriminant Analysis. *J. Am. Stat. Assoc.* 84:165–175. <https://doi.org/10.1080/01621459.1989.10478752>.
- Goutte, C., and E. Gaussier. 2005. A Probabilistic Interpretation of Precision, Recall and F-Score, with Implication for Evaluation. Pages 345–359 in *Advances in Information Retrieval*. Springer, Berlin, Heidelberg.
- Green, L. E., J. N. Huxley, C. Banks, and M. J. Green. 2014. Temporal associations between low body condition, lameness and milk yield in a UK dairy herd. *Prev. Vet. Med.* 113:63–71. <https://doi.org/10.1016/j.prevetmed.2013.10.009>.
- Groeneveld, M., D. C. J. Main, D. Tisdall, T. G. Knowles, and N. J. Bell. 2014. Measuring the response to therapeutic foot trimming in dairy cows with fortnightly lameness scoring. *Vet. J.* 201:283–288. <https://doi.org/10.1016/j.tvjl.2014.05.017>.
- Hastie, T., R. Tibshirani, and J. Friedman. 2009. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Second Edition. Springer Series in Statistics. Springer, New York, NY.
- Head, M. L., L. Holman, R. Lanfear, A. T. Kahn, and M. D. Jennions. 2015. The extent and consequences of p-hacking in science. *PLoS Biol.* 13:e1002106. <https://doi.org/10.1371/journal.pbio.1002106>.
- Hernandez, J. A., E. J. Garbarino, J. K. Shearer, C. A. Risco, and W. W. Thatcher. 2007. Evaluation of the efficacy of prophylactic hoof health examination and trimming during midlactation in reducing the incidence of lameness during late lactation in dairy cows. *J. Am. Vet. Med. Assoc.* 230:89–93. <https://doi.org/10.2460/javma.230.1.89>.
- Leach, K. A., D. A. Tisdall, N. J. Bell, D. C. J. Main, and L. E. Green. 2012. The effects of early treatment for hindlimb lameness in dairy cows on four commercial UK farms. *Vet. J.* 193:626–632. <https://doi.org/10.1016/j.tvjl.2012.06.043>.
- Leach, K. A., H. R. Whay, C. M. Maggs, Z. E. Barker, E. S. Paul, A. K. Bell, and D. C. J. Main. 2010. Working towards a reduction in cattle lameness: 1. Understanding barriers to lameness control on dairy farms. *Res. Vet. Sci.* 89:311–317. <https://doi.org/10.1016/j.rvsc.2010.02.014>.
- Lim, P. Y., J. N. Huxley, J. A. Willshire, M. J. Green, A. R. Othman, and J. Kaler. 2015. Unravelling the temporal association between lameness and body condition score in dairy cattle using a multistate modelling approach. *Prev. Vet. Med.* 118:370–377. <https://doi.org/10.1016/j.prevetmed.2014.12.015>.
- Logan, F., C. G. McAloon, E. G. Ryan, L. O'Grady, M. Duane, B. Deane, and C. I. McAloon. 2024. Sensitivity and specificity of mobility scoring for the detection of foot lesions in pasture-based Irish dairy cows. *J. Dairy Sci.* 107:3197–3206. <https://doi.org/10.3168/jds.2023-23928>.
- Mostert, P. F., C. E. van Middelaar, I. J. M. de Boer, and E. A. M. Bokkers. 2018. The impact of foot lesions in dairy cows on greenhouse gas emissions of milk production. *Agric. Syst.* 167:206–212. <https://doi.org/10.1016/j.agsy.2018.09.006>.
- O'Brien, K. M., K. G. Lawrence, and A. P. Keil. 2022. The case for case-cohort: An applied epidemiologist's guide to re-framing case-cohort studies to improve usability and flexibility. *Epidemiology* 33:354–361. <https://doi.org/10.1097/EDE.0000000000001469>.
- O'Leary, N. W., D. T. Byrne, A. H. O'Connor, and L. Shalloo. 2020. Invited review: Cattle lameness detection with accelerometers. *J. Dairy Sci.* 103:3895–3911. <https://doi.org/10.3168/jds.2019-17123>.
- Ózsvári, L. 2017. Economic cost of lameness in dairy cattle herds. *J. Dairy Vet. Anim. Res.* 6:00176. <https://doi.org/10.15406/jdvar.2017.06.00176>.
- Posit team. 2024. RStudio: Integrated Development Environment for R. Accessed Jul. 1, 2025. <https://posit.co>.
- Puerto, M. A., E. Shepley, R. I. Cue, D. Warner, J. Dubuc, and E. Vasseur. 2021. The hidden cost of disease: II. Impact of the first incidence of

- lameness on the production and economic indicators of primiparous dairy cows. *J. Dairy Sci.* 104:7944–7955. <https://doi.org/10.3168/jds.2020-19585>.
- R Core Team. 2024. R: A Language and Environment for Statistical Computing. Accessed Jul. 1, 2025. <https://www.r-project.org/>.
- Randall, L. V., M. J. Green, L. E. Green, M. G. G. Chagunda, C. Mason, S. C. Archer, and J. N. Huxley. 2018. The contribution of previous lameness events and body condition score to the occurrence of lameness in dairy herds: A study of 2 herds. *J. Dairy Sci.* 101:1311–1324. <https://doi.org/10.3168/jds.2017-13439>.
- Roever, C., N. Raabe, K. Luebke, U. Ligges, G. Szeppannek, M. Zentgraf, and D. Meyer. 2023. *klaR: Classification and Visualization*.
- Saito, T., and M. Rehmsmeier. 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS One* 10:e0118432. <https://doi.org/10.1371/journal.pone.0118432>.
- Sarker, I. H. 2021. Machine learning: Algorithms, real-world applications and research directions. *SN Comput. Sci.* 2:160. <https://doi.org/10.1007/s42979-021-00592-x>.
- Shearer, J., D. Anderson, W. Ayars, E. Belknap, S. Berry, C. Guard, K. Hoblet, E. Hovingh, G. Kirksey, A. Langill, A. Mills, D. Miskimins, J. Osterstock, R. Price, D. Prigel, A. Roussel, S. van Amstel, R. Wallace, J. Wasson, N. Cook, E. F. Garrett, D. E. Hostetler, and L. Schugel. 2004. A recordkeeping system for capture of lameness and foot-care information in cattle. *Bov. Pract.* 38:83–92. <https://doi.org/10.21423/bovine-vol38no1p83-92>.
- Siachos, N., B. E. Griffiths, J. P. Wilson, C. Bedford, A. Anagnostopoulos, J. M. Neary, R. F. Smith, and G. Oikonomou. 2025. Evaluation of a fully automated 2-dimensional imaging system for real-time cattle lameness detection using machine learning. *J. Dairy Sci.* 108:4206–4224. <https://doi.org/10.3168/jds.2024-25940>.
- Siachos, N., M. Lennox, A. Anagnostopoulos, B. E. Griffiths, J. M. Neary, R. F. Smith, and G. Oikonomou. 2024. Development and validation of a fully automated 2-dimensional imaging system generating body condition scores for dairy cows using machine learning. *J. Dairy Sci.* 107:2499–2511. <https://doi.org/10.3168/jds.2023-23894>.
- Solano, L., H. W. Barkema, S. Mason, E. A. Pajor, S. J. LeBlanc, and K. Orsel. 2016. Prevalence and distribution of foot lesions in dairy cattle in Alberta, Canada. *J. Dairy Sci.* 99:6828–6841. <https://doi.org/10.3168/jds.2016-10941>.
- Stoddard, G., N. Cook, S. Wagner, L. Solano, E. Shepley, and G. Cramer. 2025. A randomized controlled trial to evaluate the effect of 2 hoof trimming methods at dry-off on hoof lesion and lameness occurrence in dairy cattle. *J. Dairy Sci.* 108:5244–5256. <https://doi.org/10.3168/jds.2024-26068>.
- Stoddard, G., and G. Cramer. 2024. Observational study evaluating the association of hoof trimming with dairy cattle behavior and milk yield in Canada and the United Kingdom. *JDS Commun.* 5:322–326. <https://doi.org/10.3168/jdsc.2023-0493>.
- Swartz, D., E. Shepley, and G. Cramer. 2025. Evaluating cow identification reliability of a camera-based locomotion and body condition scoring system in dairy cows. *JDS Commun.* 6:202–205. <https://doi.org/10.3168/jdsc.2024-0659>.
- Swartz, D., and G. Cramer. 2026. Hoof lesion detection in lactating Holsteins: Part I. External validation of autonomous camera-derived locomotion score thresholds. *J. Dairy Sci.* 7070–7081. <https://doi.org/10.3168/jds.2025-27831>.
- Swartz, D., E. Shepley, K. P. Gaddis, J. Burchard, and G. Cramer. 2024. Descriptive evaluation of a camera-based dairy cattle lameness detection technology. *J. Dairy Sci.* 107:9847–9861. <https://doi.org/10.3168/jds.2024-24851>.
- Swartz, D. M. 2025. Effectiveness and Producers Perceptions of Camera-Based Technology Detecting Hoof Lesions in Dairy Cows. Ph.D. thesis. Veterinary Medicine, University of Minnesota, St. Paul, Mn.
- Thomas, H. J., G. G. Miguel-Pacheco, N. J. Bollard, S. C. Archer, N. J. Bell, C. Mason, O. J. R. Maxwell, J. G. Remnant, P. Sleeman, H. R. Whay, and J. N. Huxley. 2015. Evaluation of treatments for claw horn lesions in dairy cows in a randomized controlled trial. *J. Dairy Sci.* 98:4477–4486. <https://doi.org/10.3168/jds.2014-8982>.
- Thomas, H. J., J. G. Remnant, N. J. Bollard, A. Burrows, H. R. Whay, N. J. Bell, C. Mason, and J. N. Huxley. 2016. Recovery of chronically lame dairy cows following treatment for claw horn lesions: A randomised controlled trial. *Vet. Rec.* 178:116. <https://doi.org/10.1136/vr.103394>.
- USDA. 2016. Dairy 2014: Dairy Cattle Management Practices in the United States, 2014. USDA–Animal and Plant Health Inspection Service (APHIS), Veterinary Services (VS), Center for Epidemiology and Animal Health (CEAH), Fort Collins, CO.
- Van Hertem, T., Y. Parmet, M. Steensels, E. Maltz, A. Antler, A. A. Schlageter-Tello, C. Lokhorst, C. E. B. Romanini, S. Viazzi, C. Bahr, D. Berckmans, and I. Halachmi. 2014. The effect of routine hoof trimming on locomotion score, ruminating time, activity, and milk yield of dairy cows. *J. Dairy Sci.* 97:4852–4863. <https://doi.org/10.3168/jds.2013-7576>.
- van Huyssteen, M., H. W. Barkema, S. Mason, and K. Orsel. 2020. Association between lameness risk assessment and lameness and foot lesion prevalence on dairy farms in Alberta, Canada. *J. Dairy Sci.* 103:11750–11761. <https://doi.org/10.3168/jds.2019-17819>.
- Van Nuffel, A., I. Zwervaegher, S. Van Weyenberg, M. Pastell, V. M. Thorup, C. Bahr, B. Sonck, and W. Saeys. 2015. Lameness Detection in Dairy Cows: Part 2. Use of Sensors to Automatically Register Changes in Locomotion or Behavior. *Animals (Basel)* 5:861–885. <https://doi.org/10.3390/ani5030388>.
- von Keyserlingk, M. A. G., A. Barrientos, K. Ito, E. Galo, and D. M. Weary. 2012. Benchmarking cow comfort on North American freestall dairies: Lameness, leg injuries, lying time, facility design, and management for high-producing Holstein dairy cows. *J. Dairy Sci.* 95:7399–7408. <https://doi.org/10.3168/jds.2012-5807>.
- Whay, H. R., and J. K. Shearer. 2017. The Impact of Lameness on Welfare of the Dairy Cow. *Vet. Clin. North Am. Food Anim. Pract.* 33:153–164. <https://doi.org/10.1016/j.cvfa.2017.02.008>.
- Whay, H. R., A. E. Waterman, A. J. F. Webster, and J. K. O'Brien. 1998. The influence of lesion type on the duration of hyperalgesia associated with hindlimb lameness in dairy cattle. *Vet. J.* 156:23–29. [https://doi.org/10.1016/S1090-0233\(98\)80058-0](https://doi.org/10.1016/S1090-0233(98)80058-0).
- Whittle, R., J. Ensor, L. Archer, G. S. Collins, P. Dhiman, A. Denniston, J. Alderman, A. Legha, M. van Smeden, K. G. Moons, J.-B. Cazier, R. D. Riley, and K. I. E. Snell. 2025. Extended sample size calculations for evaluation of prediction models using a threshold for classification. *BMC Med. Res. Methodol.* 25:170. <https://doi.org/10.1186/s12874-025-02592-4>.
- Wilson, J. P., M. J. Green, L. V. Randall, C. S. Rutland, N. J. Bell, H. Hemingway-Arnold, J. S. Thompson, N. J. Bollard, and J. N. Huxley. 2022. Effects of routine treatment with nonsteroidal anti-inflammatory drugs at calving and when lame on the future probability of lameness and culling in dairy cows: A randomized controlled trial. *J. Dairy Sci.* 105:6041–6054. <https://doi.org/10.3168/jds.2021-21329>.
- Wynands, E. M., L. S. Caixeta, and G. Cramer. 2022. Using a target trial approach to evaluate the role of hyperketonemia in sole ulcer and white line hoof lesion development. *J. Dairy Sci.* 105:6164–6174. <https://doi.org/10.3168/jds.2021-21694>.

ORCIDS

D. Swartz, <https://orcid.org/0000-0003-3619-7055>
 A. Rendahl, <https://orcid.org/0000-0001-5434-3592>
 G. Cramer <https://orcid.org/0000-0003-2691-3417>